2. An element is composed of _____ atoms.
3. An atom is made up of _____, _____ and _____.
4. A negatively charged ion is called _____, while positively charged ion is called _____
5. _____ is a negatively charged particle (Electron/Proton).
6. Proton is deflected towards the _____ charged plate (positively, negatively).

III. Match the following.

Law of conservation of mass	Sir William Crookes
Law of constant proportion	James Chadwick
Cathode rays	Joseph Proust
Anode rays	Lavoisier
Neutrons	Goldstein

IV. Answer briefly.

1. State the law of conservation of mass.
2. State the law of constant proportions.
3. Write the properties of anode rays.
4. Define valency of an element with respect to hydrogen.
5. Define the term ions or radicals.
6. What is a chemical equation?
7. Write the names of the following compounds.

   a) CO b) N_2O c) NO_2 d) PCl_5

V. Answer the following.

1. Find the valency of the element which is underlined in the following formula.

   a) $\underline{Na}Cl$ b) $\underline{C}O_2$ c) $\underline{Al}(PO_4)$

   d) $\underline{Ba}(NO_3)_2$ e) $\underline{Ca}Cl_2$

2. Write the chemical formula for the following compounds

   a) Aluminium sulphate b) Silver nitrate

   c) Magnesium oxide d) Barium chloride

3. Write the skeleton equation for the following word equation and then balance them.

a) Carbon + Oxygen → Carbon dioxide

b) Phosphorus + Chlorine → Phosphorus pentachloride.

c) Sulphur + Oxygen → Sulphur dioxide

d) Magnesium + hydrogen chloride → Magnesium chloride + Hydrogen

4. Balance the following chemical equation.

   a) $Na + O_2 \rightarrow Na_2O$

   b) $Ca + N_2 \rightarrow Ca_3N_2$

   c) $N_2 + H_2 \rightarrow NH_3$

   d) $CaCO_3 + HCl \rightarrow CaCl_2 + CO_2 + H_2O$

   e) $Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$

VI. Higher Order Thinking Questions.

1. Why does a light paddle wheel placed in the path of cathode rays begin to rotate, when cathode rays fall on it?
2. How can we prove that the electrons carry negative charge?
3. Ruthresh, Hari, Kanishka and Thahera collected different samples of water from a well, a pond, a river and underground water. All these samples were sent to a testing laboratory. The test result showed the ratio of hydrogen to oxygen as 1:8.

   a) What conclusion would you draw from the above experiment?

   b) Which law of chemical combination does it obey?

REFERENCE BOOKS

1. Petrucci, Ralph H et.al. General Chemistry: Principles & Modern Applications (9th Edition). Upper Saddle River, NJ: Pearson Prentice Hall, 2007. Print.

8th_Science_Unit-12_EM.indd 137 21-02-2020 17:01:51